

Research Report

Query: LORA qLORA LLM fine-tuning techniques

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Executive Summary

- This report synthesizes recent advancements in fine-tuning techniques for Large Language Models (LLMs), focusing on Low-Rank Adaptation (LoRA) and its variant, QLoRA. These methods significantly reduce computational costs and memory requirements while maintaining model performance, making them accessible for broader applications.

Key Points

- LoRA freezes pre-trained model weights and fine-tunes small low-rank matrices, reducing the number of trainable parameters by up to 90%.
- QLoRA combines 4-bit quantization with LoRA, enabling fine-tuning of large models on consumer-grade hardware.

Important Findings

- QLoRA can achieve similar performance to full fine-tuning while requiring significantly less memory and computational resources.
- Parameter-efficient fine-tuning techniques like LoRA and QLoRA can outperform traditional full fine-tuning methods in certain scenarios, avoiding issues like catastrophic forgetting.

Actionable Insights

- When implementing LoRA, ensure it is applied across all layers for optimal performance.
- Adjusting hyperparameters such as LoRA rank and alpha value is crucial for maximizing model efficiency and effectiveness.

References

- Are You Still Using LoRA to Fine-Tune Your LLM?
<https://towardsdatascience.com/are-you-still-using-lora-to-fine-tune-your-llm>
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- QLoRA: Efficient Fine-Tuning of Quantized Language Models
<https://mbrenndoerfer.com/writing/qlora-efficient-finetuning-quantized-language-models>
- Efficient Fine-Tuning with LoRA for LLMs <https://www.databricks.com/blog/efficient-fine-tuning-lora-guide-llms>
- Fine-tuning Large Language Models (LLMs) Using QLoRA
<https://www.geeksforgeeks.org/nlp/fine-tuning-large-language-models-llms-using-qlora>